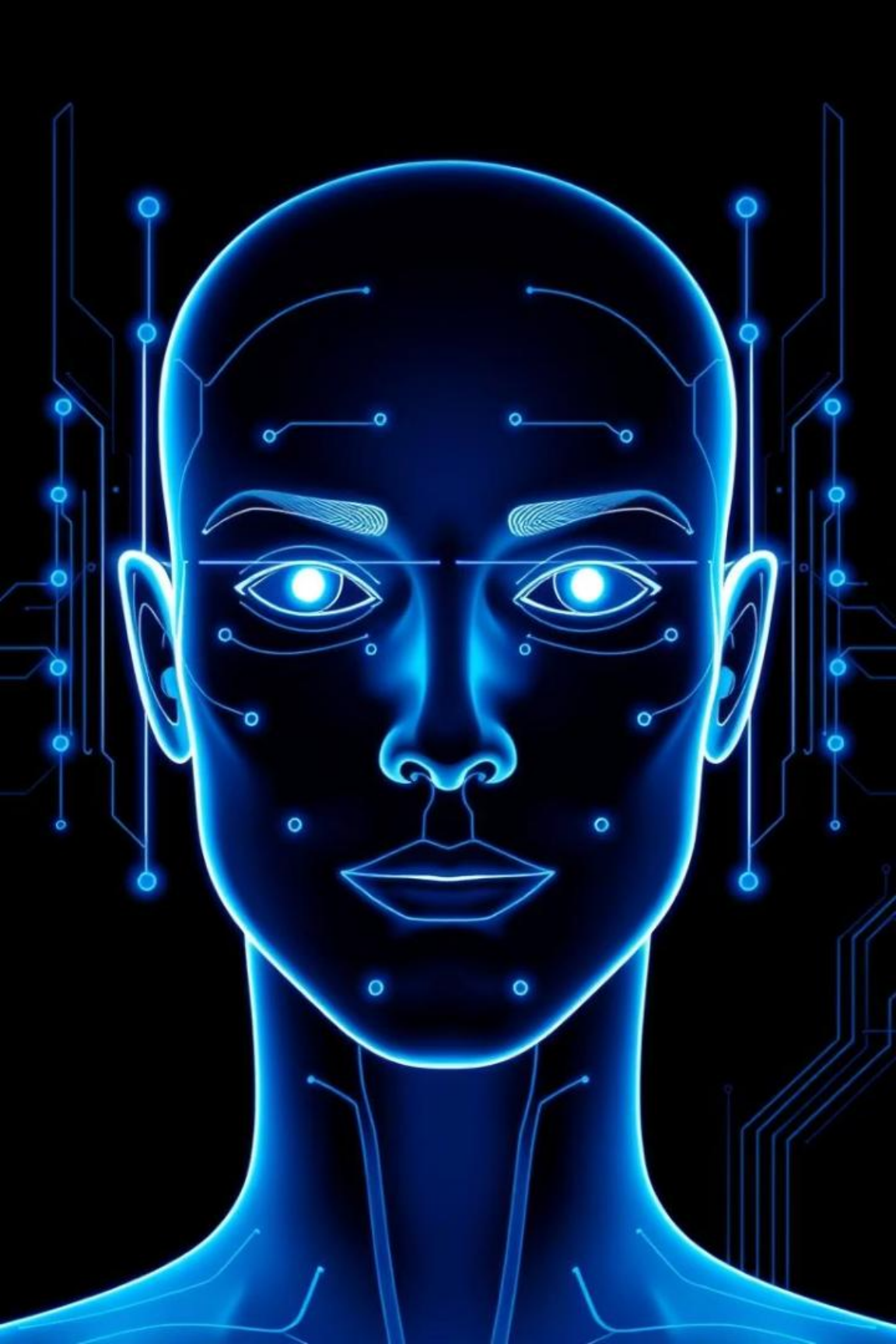




# Face Recognition Attendance System

Automating Attendance Tracking with AI

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# Problem Statement

## Manual System Issues

- Time-consuming roll calls delay classes
- Proxy attendance by friends
- Human errors and lost records

## Our AI Solution

- AI-based facial recognition
- Zero physical contact, post-pandemic safe

# Key Objectives

## Automate Attendance

Real-time face detection replaces paper

## High Accuracy

Over 85% recognition rate with LBPH

## Prevent Fraud

Biometric verification stops proxy attendance

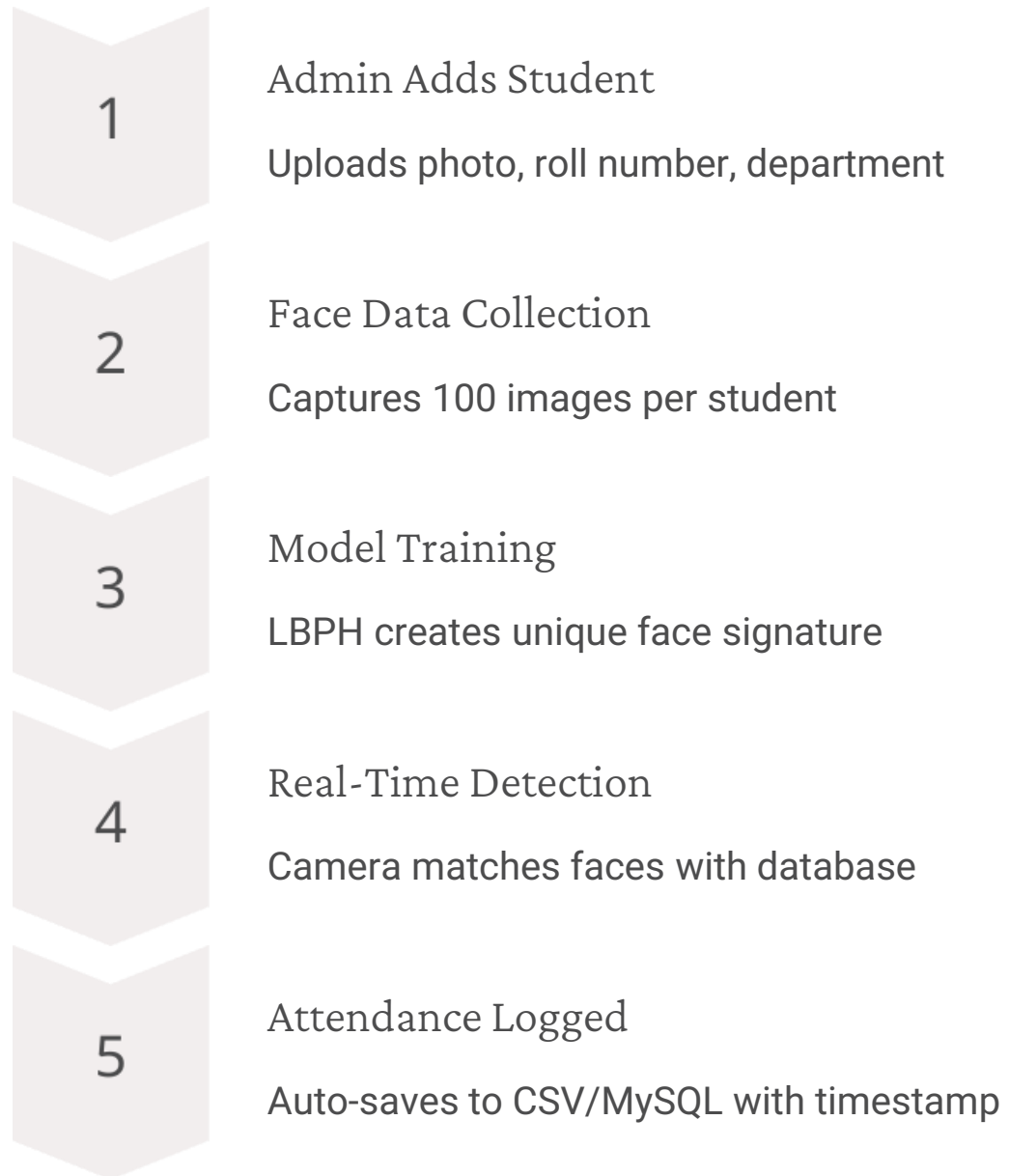
## User-Friendly Interface

Simple GUI to add students and view logs

## Scalability

Supports 1000+ students with MySQL

# System Workflow



# Technical Stack

| Face Detection      | Face Recognition                      | Database                               | GUI                        |
|---------------------|---------------------------------------|----------------------------------------|----------------------------|
| OpenCV Haar Cascade | LBPH Algorithm: lightweight, accurate | MySQL stores roll numbers, departments | Tkinter, Python-compatible |
| Backend             |                                       |                                        |                            |
| Python + OpenCV     |                                       |                                        |                            |

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# Code Highlights

## Face Detection

detectMultiScale for real-time face detection

## Database Integration

Insert student data into MySQL

## Attendance Logging

Append attendance with timestamp to CSV

# Dataset & Training

## Dataset Structure

Folder: data/ with user.{rollno}.{id}.jpg files

## Training Process

- Convert images to grayscale
- Extract LBPH features
- Save model as classifier.xml

# Results & Performance

- Accuracy: 85-90% in good lighting
- Speed: 10-15 faces/second at 720p
- Storage: MySQL handles 10,000+ records
- CSV for daily backups

| Metric       | Traditional | Our System |
|--------------|-------------|------------|
| Time/Student | 30 sec      | 2 sec      |
| Proxy Risk   | High        | None       |



# Challenges & Fixes

| Challenge             | Solution                     |
|-----------------------|------------------------------|
| Low-light recognition | Adaptive thresholding added  |
| False positives       | Increased training samples   |
| Database lag          | Batch processing implemented |



# Test Metrics

## Face Detection Report:

- Precision: 0.89
- Recall: 0.92
- F1-Score: 0.90

LBPH Recognition Accuracy: 87.5%

FPS: 12.3

Noise Robustness: 8% drop (Gaussian)

Database Query: 0.32 sec

# Future Scope

- 1 — 2024: Mobile App  
Android/iOS attendance marking
- 2 — 2025: Cloud Sync  
Live logs on AWS/Azure
- 3 — Deep Learning Upgrade  
CNN for 95%+ accuracy
- 4 — Multi-Factor Auth  
Face + RFID/NFC

